Develop a "Student Utility Program" that performs the following operations using a menu-driven approach.Part A: Java Basics

* Display student information.
* Accept Name, Roll Number, and Marks.
* Calculate Total and Percentage.
* Display the result.

Steps:
Start the program.
Import the Scanner class for user input.
Create methods for factorial, prime number, maximum, and area of circle.
Create a Scanner object.
Display the Student Utility menu.
Read the user's choice.
Use a switch statement to perform the selected operation.
Accept student details and calculate total and percentage.
Perform conditional operations like even/odd, largest number, grade, and day.
Perform looping operations like multiplication table, numbers from 1 to N, sum, and Fibonacci series.
Call methods for factorial, prime, maximum, and circle area.
Display the result.
Repeat the menu until the user selects Exit (14).
Close the Scanner.

Algorithm:
Step 1: Start.
Step 2: Import Scanner and create a Scanner object.
Step 3: Define methods:
factorial(n) → calculates factorial.
prime(n) → checks whether a number is prime.
maximum(a,b) → finds maximum of two numbers.
areaCircle(r) → calculates area of circle.
Step 4: Display the menu from 1 to 14.
Step 5: Read the user's choice.
Step 6: Execute the corresponding operation using switch-case.
Step 7: Display the result.
Step 8: If choice is 14, display "Thank You!" and exit.
Step 9: Otherwise, return to the menu.
Step 10: Close Scanner and stop.

UML Diagram:

```
 +--------------------------------------+
|       StudentUtilityProgram          |
```

```
+--------------------------------------+
|                                      |
+--------------------------------------+
| + factorial(n : int) : int           |
| + prime(n : int) : boolean           |
| + maximum(a : int,b : int) : int     |
| + areaCircle(r : double) : double    |
| + main(args : String[]) : void       |
+--------------------------------------+
              |
              | uses
              v
+--------------------------------------+
|            Scanner                   |
+--------------------------------------+
|                                      |
+--------------------------------------+
| + nextInt() : int                    |
| + nextDouble() : double              |
| + nextLine() : String                |
| + close() : void                     |
+--------------------------------------+
```

Code:

```
 import java.util.Scanner;

public class StudentUtilityProgram {


    public static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    public static boolean prime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i < n; i++) {
            if (n % i == 0)
                return false;
```

```
    }
    return true;
}

public static int maximum(int a, int b) {
    if (a > b)
        return a;
    else
        return b;
}

public static double areaCircle(double r) {
    return 3.14 * r * r;
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);
    int choice;

    do {

        System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
        System.out.println("1. Student Information");
        System.out.println("2. Even or Odd");
        System.out.println("3. Largest of Three Numbers");
        System.out.println("4. Grade");
        System.out.println("5. Day of the Week");
        System.out.println("6. Multiplication Table");
        System.out.println("7. Numbers from 1 to N");
        System.out.println("8. Sum of First N Natural Numbers");
        System.out.println("9. Fibonacci Series");
        System.out.println("10. Factorial");
        System.out.println("11. Prime Number");
        System.out.println("12. Maximum of Two Numbers");
        System.out.println("13. Area of Circle");
        System.out.println("14. Exit");
        System.out.print("Enter your choice: ");

        choice = sc.nextInt();

        switch (choice) {
```

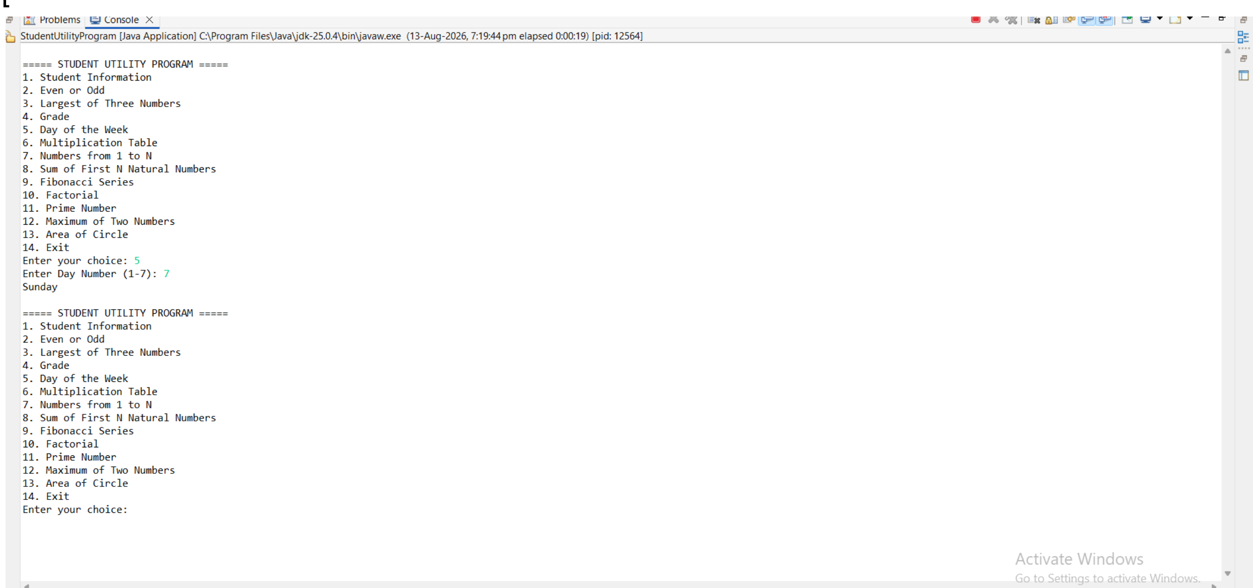
```
case 1:
    sc.nextLine();

    System.out.print("Enter Student Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Roll Number: ");
    int roll = sc.nextInt();

    System.out.print("Enter Marks in Subject 1: ");
    int m1 = sc.nextInt();

    System.out.print("Enter Marks in Subject 2: ");
    int m2 = sc.nextInt();

    System.out.print("Enter Marks in Subject 3: ");
    int m3 = sc.nextInt();

    int total = m1 + m2 + m3;
    double percentage = total / 3.0;

    System.out.println("\nStudent Name : " + name);
    System.out.println("Roll Number  : " + roll);
    System.out.println("Total Marks  : " + total);
    System.out.println("Percentage   : " + percentage);
    break;


case 2:
    System.out.print("Enter Number: ");
    int num = sc.nextInt();

    if (num % 2 == 0)
        System.out.println("Even Number");
    else
        System.out.println("Odd Number");
    break;

case 3:
    System.out.print("Enter First Number: ");
    int a = sc.nextInt();

    System.out.print("Enter Second Number: ");
    int b = sc.nextInt();
```

```
    System.out.print("Enter Third Number: ");
    int c = sc.nextInt();

    if (a >= b && a >= c)
        System.out.println("Largest = " + a);
    else if (b >= a && b >= c)
        System.out.println("Largest = " + b);
    else
        System.out.println("Largest = " + c);
    break;

case 4:
    System.out.print("Enter Percentage: ");
    double per = sc.nextDouble();

    if (per >= 90)
        System.out.println("Grade A");
    else if (per >= 75)
        System.out.println("Grade B");
    else if (per >= 60)
        System.out.println("Grade C");
    else if (per >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Fail");
    break;

case 5:
    System.out.print("Enter Day Number (1-7): ");
    int day = sc.nextInt();

    switch (day) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    case 3:
        System.out.println("Wednesday");
        break;
    case 4:
        System.out.println("Thursday");
```

```
        break;
    case 5:
        System.out.println("Friday");
        break;
    case 6:
        System.out.println("Saturday");
        break;
    case 7:
        System.out.println("Sunday");
        break;
    default:
        System.out.println("Invalid Day");
    }
    break;


case 6:
    System.out.print("Enter Number: ");
    int table = sc.nextInt();

    for (int i = 1; i <= 10; i++) {
        System.out.println(table + " x " + i + " = " + (table * i));
    }
    break;

case 7:
    System.out.print("Enter N: ");
    int n = sc.nextInt();

    for (int i = 1; i <= n; i++) {
        System.out.print(i + " ");
    }
    System.out.println();
    break;

case 8:
    System.out.print("Enter N: ");
    int num1 = sc.nextInt();

    int sum = 0;
    for (int i = 1; i <= num1; i++) {
        sum += i;
    }
```

```java
    System.out.println("Sum = " + sum);
    break;

case 9:
    System.out.print("Enter Number of Terms: ");
    int terms = sc.nextInt();

    int f1 = 0, f2 = 1;

    System.out.print("Fibonacci Series: ");

    for (int i = 1; i <= terms; i++) {
        System.out.print(f1 + " ");
        int next = f1 + f2;
        f1 = f2;
        f2 = next;
    }
    System.out.println();
    break;


case 10:
    System.out.print("Enter Number: ");
    int fact = sc.nextInt();

    System.out.println("Factorial = " + factorial(fact));
    break;

case 11:
    System.out.print("Enter Number: ");
    int prime = sc.nextInt();

    if (prime(prime))
        System.out.println("Prime Number");
    else
        System.out.println("Not a Prime Number");
    break;

case 12:
    System.out.print("Enter First Number: ");
    int x = sc.nextInt();

    System.out.print("Enter Second Number: ");
    int y = sc.nextInt();
```

```
                System.out.println("Maximum = " + maximum(x, y));
                break;

            case 13:
                System.out.print("Enter Radius: ");
                double r = sc.nextDouble();

                System.out.println("Area = " + areaCircle(r));
                break;

            case 14:
                System.out.println("Thank You!");
                break;

            default:
                System.out.println("Invalid Choice");
            }

        } while (choice != 14);

        sc.close();
    }
}
[
```


Problems Console
StudentUtilityProgram [Java Application] C:\Program Files\Java\jdk-25.0.4\bin\javaw.exe (13-Aug-2026, 7:19:44 pm elapsed 0:00:19) [pid: 12564]

```
===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade
5. Day of the Week
6. Multiplication Table
7. Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 5
Enter Day Number (1-7): 7
Sunday

===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade
5. Day of the Week
6. Multiplication Table
7. Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice:
```